



Cislunar Space Situational Awareness

Intelligence & Analytics for the New Space Domain

Preliminary Business Plan · March 2026

Vision

Become the definitive intelligence layer for cislunar space situational awareness — the domain nobody is watching yet, that every lunar operator will need within five years.

Why This Works

Starts with free public data and standard engineering tools. No proprietary hardware required to begin generating value. Cislunar is underserved by incumbents. The market is forming now.

1. Executive Summary

CislunarWatch is a space situational awareness (SSA) analytics company focused exclusively on the cislunar domain — the volume of space between Earth and the Moon, including lunar orbit. The company aggregates publicly available and licensed tracking data, applies advanced orbital mechanics modeling, and delivers actionable intelligence products to satellite operators, government agencies, launch providers, insurance underwriters, and the growing community of lunar mission operators.

THE CORE INSIGHT

Every existing SSA company is built around Low Earth Orbit. The cislunar domain — where all lunar activity happens — is functionally unwatched commercially. As Artemis missions multiply, commercial landers proliferate, and lunar orbital stations are built, the collision risk and traffic management problem will become acute. CislunarWatch establishes domain expertise before the market is crowded, not after.

Metric	Detail
--------	--------

Company Name	CislunarWatch (working name)
Business Model	Data analytics + intelligence products as a service
Starting Point	Free public data (Space-Track.org) + open source orbital mechanics libraries
First Customer Target	NASA Artemis mission planners, commercial lunar lander operators
Market	\$11.4B+ lunar exploration tech; \$1.2B+ commercial SSA market growing rapidly
Capital to Start	Near zero for initial R&D; seed round \$500K-\$2M for product build
Competitive Position	First mover in underserved domain; expertise built before competition arrives

2. The Problem

Orbital Space Is Getting Crowded — Especially Near the Moon

Low Earth Orbit has a well-documented debris and congestion problem with established commercial solutions. The cislunar domain — everything beyond GEO out to lunar distance and including lunar orbit — is largely unmonitored commercially, and activity there is about to increase dramatically.

- NASA Artemis program will put humans at the lunar south pole with multiple missions, multiple vehicles, orbital rendezvous, and Gateway station in NRHO orbit.
- Commercial landers: Intuitive Machines, Astrobotic, Firefly, ispace — multiple missions per year planned through 2030.
- China's lunar program: crewed south pole mission before 2030 with independent orbital infrastructure.
- SpaceX Starship lunar cargo deliveries targeted 2027-2028 — large vehicles in cislunar space on a recurring basis.
- Proposed lunar orbital stations, communication relay satellites, and navigation beacons — all adding objects requiring tracking.

THE GAP

The US Space Force's 18th Space Control Squadron tracks objects in LEO and GEO. Cislunar space has no equivalent commercial monitoring infrastructure. The orbital mechanics are genuinely more complex — three-body problem dynamics, lunar gravity perturbations, chaotic trajectories near Lagrange points. Existing LEO tools do not transfer cleanly. No company has built cislunar SSA as a dedicated commercial product.

Why This Matters

A collision in cislunar space or lunar orbit is categorically different from a LEO collision. There is no atmospheric drag to clean up debris over time. Objects in lunar orbit or at Lagrange points can remain there for centuries. A fragmentation event near the Moon could contaminate the entire lunar orbital environment permanently — degrading humanity's ability to operate there for generations. The stakes are civilizational, not merely commercial.

3. The Opportunity

A Domain Nobody Owns Yet

The commercial SSA market is real and growing. LeoLabs, ExoAnalytic, and AGI (with STK software) are established players — all fundamentally LEO-oriented. Their sensor networks, orbital models, and customer relationships are optimized for the LEO environment.

Cislunar SSA has different requirements:

- Different orbital mechanics: three-body dynamics, lunar gravity harmonics, halo orbits, libration point stability — not well served by LEO propagators.
- Different sensor geometry: ground-based radar covering LEO does not see deep cislunar space; different telescope networks and tracking cadences are required.
- Different customers: lunar mission operators, Gateway program, Artemis mission planners — not the LEO satellite constellation operators that incumbents serve.
- Different regulatory context: no established cislunar traffic management framework exists — the company that builds the intelligence layer helps define the standards.

TIMING

The cislunar market is where LEO SSA was in 2010 — real activity beginning, no established commercial intelligence infrastructure, window open for a first mover to define the space. LeoLabs was founded in 2016 and is now valued in the hundreds of millions. The cislunar equivalent founded in 2026 has the same trajectory available.

4. Starting Without Proprietary Hardware

The Data Already Exists

The critical structural insight: the data layer and the sensor layer are separable. An intelligence and analytics business can be built first, with proprietary sensing added later once revenue justifies the investment.

Free Public Data Sources

Source	What It Provides
Space-Track.org (free)	Two-Line Element sets for ~27,000 tracked objects; updated multiple times daily; free registration
NASA JPL Horizons	Precise ephemeris data for all solar system bodies including Moon, Lagrange points, and spacecraft; free API
NASA SPICE kernels	Spacecraft trajectory data for past and current missions; free download
ESA DISCOS database	Launch history, object characteristics, fragmentation records; free registration
NASA Orbital Debris Program	Research datasets, size distribution models, collision probability tools; public access
Celestrak.org	Curated TLE datasets and supplemental data; free and actively maintained
Space Fence public products	US Space Force publishes conjunction data messages (CDMs) for close approaches; free

Licensable Data (Phase 2)

Source	What It Adds
LeoLabs API	Higher precision radar tracking; reduces position uncertainty significantly
ExoAnalytic	Optical tracking data; GEO and beyond; fills gaps in radar coverage

Open Source Software Stack

The orbital mechanics software ecosystem is mature, free, and well-documented:

```
# Core Python libraries
pip install skyfield           # High-precision astronomy; JPL ephemerides
```

```
pip install sgp4           # Standard TLE propagator; satellite
position from TLEs
pip install poliastro      # Orbital mechanics; cislunar trajectory
modeling
pip install astropy        # Astronomy calculations; coordinate
transforms
pip install plotly         # 3D visualization of orbital data
pip install pandas numpy   # Data handling

# Key data sources
# space-track.org          - Free registration; full TLE catalog
# ssd.jpl.nasa.gov/horizons - JPL ephemeris API
# celestrak.org            - Curated TLE sets
```

5. Product Roadmap

Phase 1: Analytics Foundation

The first product is demonstrated capability — analysis, visualization, and published insight that establishes CislunarWatch as the authoritative voice on cislunar SSA before competitors recognize the opportunity.

Deliverable 1: Cislunar Object Catalog

Pull all TLEs from Space-Track. Filter for objects with semi-major axes beyond GEO (~42,000 km). Visualize current positions and projected trajectories in 3D. The currently-tracked cislunar population is small — perhaps a few dozen active objects and some debris. CislunarWatch can build comprehensive knowledge of every object in this domain from day one. That depth of familiarity is the foundation of credible expertise.

Deliverable 2: Mission Conjunction Analysis

Pull trajectory data for upcoming Artemis and commercial lunar missions from NASA SPICE kernels. Calculate minimum approach distances between mission vehicles and every tracked cislunar object over mission timelines. Identify conjunctions worth flagging. This is analysis that NASA mission planners perform internally — CislunarWatch produces it independently and publishes it, establishing credibility with the exact customer base the company serves.

Deliverable 3: Published Domain Intelligence

Publish analysis publicly — technical articles, visualizations, open methodology. In a domain this new, rigorous public analysis creates the reputation that leads to customer conversations,

which lead to contracts. Researchers worth knowing in this domain: Moriba Jah (UT Austin, space sustainability), Marlon Sorge (Aerospace Corp, debris modeling), T.S. Kelso (Celestrak).

Phase 2: Commercial Intelligence Products

- Cislunar Conjunction Warning Service: automated daily reports of close approaches between tracked objects and planned mission assets. Delivered as API or structured digest.
- Mission Safety Analysis: commissioned conjunction analysis for specific missions — pre-launch trajectory review, on-orbit monitoring. High-value, high-margin consulting product.
- Cislunar Domain Awareness Dashboard: web-based visualization and analytics platform; subscription access; current object positions, predicted trajectories, risk assessments.
- Regulatory Intelligence: monitoring and analysis of emerging cislunar traffic management frameworks as a subscription product for operators needing compliance guidance.

Phase 3: Proprietary Sensing

Once revenue funds it, CislunarWatch invests in proprietary sensor capability — specifically deep space optical telescopes capable of tracking objects at lunar distances. A network of 2-3 telescopes at dark sky sites, combined with the analytics platform, creates a defensible data moat. This is a \$2-5M investment justified by the revenue base built in Phases 1 and 2.

6. Business Model

Revenue Stream	Timeline	Estimated Value
Mission Safety Analysis (consulting)	2026-2027	\$25K-\$150K per mission; high margin; funds early operations
Data & Intelligence API	2027-2028	\$5K-\$50K/month per customer; recurring; scales with customer count
Dashboard Subscriptions	2027-2028	\$1K-\$10K/month per operator; SaaS model
Government Contracts (SBIR/STTR)	2026-2027	\$300K-\$2M non-dilutive; NASA, Space Force all fund this domain
Regulatory Compliance Products	2028+	Growing market as cislunar traffic management frameworks develop

Proprietary Data Licensing	2029+	After sensor network is operational; highest margin product
-----------------------------------	-------	---

REVENUE SEQUENCING

Consulting revenue funds operations while the platform is built. Platform subscriptions create recurring revenue that justifies seed investment. Seed investment funds the sensor network. Sensor network creates proprietary data moat that justifies Series A. Each stage funds the next.

7. Market Analysis

Market Size

- Lunar exploration technology market: \$11.4B (2025) growing to \$21.4B by 2030 at 13.3% CAGR.
- Space mining market: \$1.44B (2025) projected to \$4.07B by 2034. Lunar sector accounts for 61%.
- Commercial SSA market: growing rapidly driven by megaconstellation expansion and regulatory pressure.
- Government investment: \$31B+ in projected global government space exploration spending through 2034.

Target Customers

Segment	Specific Need
NASA Artemis Program	Independent conjunction analysis for Gateway, landers, and crew vehicles in cislunar space; mission safety documentation
Commercial Lunar Operators	Intuitive Machines, Astrobotic, Firefly, ispace — all need orbital safety analysis for missions
SpaceX Starship Lunar	As Starship begins recurring lunar cargo missions, cislunar traffic awareness becomes operationally critical
Space Insurance Underwriters	AXA XL, Marsh, Lloyd's syndicates need independent risk assessment to price lunar mission policies
US Space Force / NRO	Cislunar domain awareness is a stated national security priority; commercial intelligence fills gaps
ESA / JAXA / ISRO	International agencies planning lunar operations need independent non-US-government data sources

Academic Research	Universities modeling the cislunar environment; research partnerships build credibility and data access
--------------------------	---

Competitive Landscape

CislunarWatch's primary competitive advantage is domain focus. No current player has built cislunar SSA as a dedicated commercial product.

- LeoLabs: best-in-class LEO radar network; no meaningful cislunar capability; serves a different customer base.
- ExoAnalytic: optical tracking with GEO focus; some deeper space capability but not cislunar-specialized.
- AGI / Ansys STK: dominant orbital analysis software platform; sells tools, not intelligence products; potential partner rather than competitor.
- NASA / Space Force internal: conduct their own analysis but require independent commercial augmentation; natural customers.

COMPETITIVE MOAT

Domain expertise plus proprietary data plus network effects. As more operators subscribe, the platform incorporates more trajectory data, improving models for all users. First mover in an uncontested domain with compounding data advantages is the most defensible position in any data business.

8. Milestones & Timeline

Phase	Period	Key Milestones
Phase 0: Foundation	Weeks 1-4	Space-Track registration; Python stack operational; first cislunar object visualization; initial customer discovery conversations
Phase 1: Establish	Months 1-3	Publish 3-5 technical analysis pieces; cislunar conjunction tool operational; identify technical co-founder; advisory network initiated
Phase 2: Validate	Months 3-6	First paying mission analysis contract; SBIR Phase I application filed; company incorporated

Phase 3: Product	Months 6-18	SBIR award; MVP dashboard live; first SaaS subscribers; seed round closed (\$500K-\$2M)
Phase 4: Scale	Year 2-3	Series A; first proprietary telescope operational; government contracts; full platform
Phase 5: Leadership	Year 3-5	Proprietary sensor network; industry standard platform; acquisition or IPO pathway

9. Funding Strategy

CislunarWatch is structurally unusual in that meaningful early progress requires minimal capital. This allows the company to build real credibility and initial revenue before approaching outside investors — a significantly stronger negotiating position than most hardware-dependent space companies.

- Phase 0-1: Zero external capital required. Free public data, open source tools. Output: published analysis, customer conversations, demonstrated domain expertise.
- Phase 2: Consulting revenue self-funds. First mission analysis contracts at \$25-150K cover operations without dilution.
- SBIR Phase I (\$300K non-dilutive): Apply to NASA SBIR and Space Force SBIR. Relevant topic areas: cislunar SSA, space traffic management, orbital conjunction analysis.
- SBIR Phase II (\$2M non-dilutive): Funds platform development and initial sensor feasibility work.
- Seed (\$500K-\$2M): Once platform has initial subscribers and SBIR is awarded. Target: Seraphim Capital, Space Capital, In-Q-Tel (given national security applications).
- Series A (\$10-20M): Post government contracts and proven SaaS metrics. Funds proprietary sensor network build-out.

10. Team Requirements

Role	Profile
CEO / Founder	Business architecture; government contracting navigation; physics or engineering background; relationship-driven go-to-market with

	NASA and commercial operators
Technical Co-Founder / CTO	Orbital mechanics, astrodynamics, or space systems engineering background; ideally from JPL, NASA Goddard, Aerospace Corporation, or a commercial SSA company
Advisor: Space Law	US Commercial Space Launch Competitiveness Act; cislunar resource rights; FCC and FAA licensing for space operations
Advisor: Government Procurement	Former NASA program manager or contracting officer; deep knowledge of SBIR, STTR, and how SSA procurement decisions are made
Advisor: Astrodynamics	Academic or industry researcher in cislunar orbital mechanics; provides technical credibility and access to research networks

11. Risk Assessment

Risk	Likelihood	Mitigation
Artemis / commercial lunar timeline slips	High — historical pattern	Phase 0-1 revenue from consulting does not depend on lunar delivery cadence; company builds expertise regardless
Incumbent expands into cislunar	Medium — possible but not immediate	First-mover expertise and data advantage are durable; being 2-3 years ahead in domain knowledge is hard to replicate quickly
Government internalizes capability	Medium — current policy favors commercial	International agencies and commercial operators are independent customers; not solely US government dependent
Regulatory framework delays revenue	Medium	Consulting and data licensing products generate revenue independent of specific regulatory requirements
Technical talent acquisition	Medium — competitive market	University partnerships and NASA/JPL alumni networks provide access; equity compensation and mission-driven culture are strong recruiting tools in this domain
Kessler cascade in LEO (different problem)	Low for cislunar	Validates urgency of the space safety market broadly; potentially

		accelerates customer adoption
--	--	-------------------------------

12. Technical Quick-Start

The following demonstrates how rapidly initial capability can be established using existing public resources:

Step 1: Data Access

```
# Register at space-track.org (free)
# Then pull the full TLE catalog:

import requests, json

session = requests.Session()
session.post('https://www.space-track.org/ajaxauth/login',
            data={'identity': 'YOUR_EMAIL', 'password': 'YOUR_PASSWORD'})

resp = session.get(
    'https://www.space-track.org/basicspacedata/query/class/tle_latest
    /ORDINAL/1/EPOCH/%3Enow-30/orderby/NORAD_CAT_ID/format/json')

tles = json.loads(resp.text)
print(f'Tracking {len(tles)} objects')
```

Step 2: Filter for Cislunar Objects

```
# Cislunar = semi-major axis beyond GEO (~42,164 km)
# Equivalent: mean motion below ~0.5 revs/day

cislunar_objects = []
for tle in tles:
    try:
        mean_motion = float(tle['MEAN_MOTION']) # revs per day
        if mean_motion < 0.5: # beyond GEO
            cislunar_objects.append(tle)
    except: pass

print(f'Cislunar objects tracked: {len(cislunar_objects)}')
# Result: 20-50 objects – the entire currently-tracked cislunar
population.
```

```
# This is the market CislunarWatch serves. It fits on one screen today.  
# In five years it will not.
```

Step 3: Visualize the Domain

```
from skyfield.api import load  
import plotly.graph_objects as go  
  
# Get Moon position from JPL ephemeris  
ts = load.timescale()  
eph = load('de421.bsp')  
earth, moon = eph['earth'], eph['moon']  
t = ts.now()  
moon_pos = (moon - earth).at(t).position.km  
  
# Plot: Earth at origin, Moon at moon_pos,  
# all cislunar objects scattered between.  
# This is the complete picture of what humanity  
# has placed in cislunar space – currently sparse,  
# growing rapidly, and unmonitored commercially.
```

From data access to first visualization is a matter of hours. From first visualization to publishable conjunction analysis is a matter of days. The gap between concept and demonstrated capability in this business is unusually small — a significant structural advantage over hardware-dependent competitors.

Summary

CislunarWatch addresses a genuine and growing gap in the space safety infrastructure — one that becomes more critical with every lunar mission added to the manifest. The business is differentiated by domain focus, can be initiated with minimal capital using existing public data, and is positioned to become the industry standard intelligence platform for cislunar operations before the domain becomes crowded.

The combination of near-term consulting revenue, non-dilutive government grant funding, and a clear path to a scalable SaaS product makes this an unusually capital-efficient entry into the space sector. The window to establish first-mover position in cislunar SSA is open now and will not remain open indefinitely.

The lunar economy is forming. The south pole is the center of gravity for all near-term activity. The safety and traffic management infrastructure for that economy does not exist yet. CislunarWatch builds it — starting from publicly available data and open source tools, scaling to proprietary sensing as revenue grows. First mover in a domain that every cislunar operator will depend on.